

SIH26082 — Product Requirements Document (PRD)

Product: High-Resolution Coupled AQI Forecasting System for Delhi NCR

PS: SIH26082 | **Category:** Software | **Primary outcome:** 72-hour high-resolution AQI forecasting with coupled meteorology–chemistry modelling

1. PRODUCT VISION

Build a forecasting and decision-support platform for Delhi NCR that models the **two-way interaction between meteorology and atmospheric pollution**. The product must forecast AQI for the next 72 hours while explicitly representing atmospheric inversion effects and regional stubble-burning plume dispersion.

The product is not defined as a conventional ML AQI predictor. Its defining technical requirement is a **coupled weather–chemistry forecasting workflow** using WRF-Chem or a similar open-source coupled framework.

2. PROBLEM STATEMENT

Traditional AQI forecasting can treat meteorology and pollution dispersion separately. In heavily polluted urban environments such as Delhi NCR, the relationship is coupled: inversion can trap pollutants near the surface, while high aerosol concentrations can alter sunlight, temperature, wind patterns and PBL height. The product must capture these interactions to improve forecast quality and provide actionable insight.

3. PRODUCT GOALS

Goal	Success condition
72-hour forecasting	Generate an AQI outlook for the next 72 hours.
Coupled modelling	Meteorology and chemistry interact dynamically in the modelling workflow.
Delhi NCR resolution	Produce spatially detailed output for the Delhi NCR domain.
Inversion awareness	Track atmospheric inversion strength and its impact on pollution.
Stubble-burning impact	Represent regional pollution spikes and predict plume dispersion.
Actionable visualization	Present forecast, drivers and spatial impact through a real-time dashboard.

4. TARGET USERS

User	Primary need
Air-quality / environmental analyst	Understand forecast evolution, pollutants and atmospheric drivers.
Research / modelling team	Inspect coupled-model outputs and validate forecast behaviour.
Decision-support user	Identify expected deterioration, hotspots and major contributing conditions.
General dashboard user	Understand where and when air quality is expected to worsen.

5. FUNCTIONAL REQUIREMENTS

ID	Requirement	Priority
FR-01	Ingest and prepare meteorological inputs including temperature, wind and PBL-related information.	Must
FR-02	Ingest/prepare pollution and chemistry inputs including PM2.5, PM10, O3 and NOx.	Must
FR-03	Integrate regional/stubble-burning emission information into the modelling workflow.	Must
FR-04	Run WRF-Chem or a similar open-source coupled weather-chemistry framework.	Must
FR-05	Generate a 72-hour forecast for Delhi NCR.	Must
FR-06	Represent two-way meteorology–chemistry feedback.	Must
FR-07	Track atmospheric inversion strength and its relationship to pollutant trapping.	Must
FR-08	Predict regional stubble-burning plume dispersion under prevailing weather.	Must
FR-09	Generate spatial AQI/pollution outputs for visualization.	Must
FR-10	Provide a real-time/user-friendly dashboard.	Must
FR-11	Expose forecast drivers and actionable interpretation.	Should

SIH26082 — PRD: System & Experience Requirements

6. CORE USER EXPERIENCE

Primary flow: User opens dashboard → sees current Delhi NCR air-quality state → selects a forecast time → views spatial AQI → inspects pollutant concentrations → examines wind/PBL/inversion conditions → views stubble-burning plume → understands expected deterioration/improvement and its major drivers.

7. DASHBOARD REQUIREMENTS

Component	Required behaviour
Delhi NCR map	Show high-resolution spatial forecast and hotspots.
72-hour timeline	Allow inspection of forecast evolution over the full horizon.
AQI layer	Display forecast AQI spatially and temporally.
Pollutant layers	Expose PM2.5, PM10, O3 and NOx information.
Meteorology	Show relevant temperature and wind information; expose PBL-related conditions.
Inversion indicator	Display inversion strength/status and associated pollution-trapping conditions.
Plume layer	Show predicted stubble-burning plume movement/impact.
Driver/explanation panel	Explain major forecast drivers such as inversion, weak ventilation and incoming plume.
Updates	Present updated model forecasts rather than a static dataset.

8. MODEL / DATA PIPELINE

Required conceptual pipeline:

Weather + pollution + emissions → coupled WRF-Chem/similar framework → meteorology/chemistry feedback → 72-hour spatial forecast → AQI generation → dashboard.

Layer	Responsibility
Data ingestion	Acquire, validate, synchronize and prepare weather, pollution and emission inputs.
Model configuration	Define Delhi NCR domain, grid, forecast horizon and relevant physics/chemistry configuration.
Coupled simulation	Run the weather-chemistry interaction and generate forecast fields.
Post-processing	Derive pollutant/AQI products, inversion indicators and plume representations.
Validation	Compare forecast outputs with observations and baseline methods.
API / service layer	Expose model outputs to the dashboard.
Dashboard	Render spatial and temporal forecast information for users.

9. NON-FUNCTIONAL REQUIREMENTS

Area	Requirement
Reliability	Missing/stale input data must be detected; failures must not silently create false forecasts.
Reproducibility	A forecast run should be reproducible from recorded inputs/configuration.
Performance	Forecast generation must fit a practical operational/demo workflow; exact runtime target is to be established during implementation.
Scalability	Architecture should permit higher-resolution grids or additional forecast domains later.
Observability	Model runs, input freshness and failures should be visible to the technical team.
Usability	Dashboard should communicate complex atmospheric information without requiring users to read raw model output.

10. VALIDATION & ACCEPTANCE

Test	Acceptance intent
Historical backtest	Run the system on historical high-pollution periods and compare against observations.
72-hour accuracy	Measure forecast performance at multiple lead times.
Spatial validation	Evaluate spatial forecast behaviour where observations permit.
Inversion event test	Verify the system captures pollution trapping during representative inversion conditions.
Stubble-burning test	Verify regional emission/plume influence is represented in the forecast.
Baseline comparison	Compare against a simpler forecasting approach to demonstrate the value of the coupled approach.
End-to-end test	Data ingestion through model run, post-processing and dashboard display works as one pipeline.

SIH26082 — PRD: Scope, MVP & Success Criteria

11. MVP SCOPE

MVP objective: demonstrate a legitimate coupled forecasting workflow and a usable Delhi NCR dashboard, without attempting every possible research enhancement.

MVP module	Scope
Forecast domain	Delhi NCR spatial domain and defined model grid.
Coupled model	Operational prototype using WRF-Chem or selected equivalent open-source coupled framework.
Inputs	Meteorological, pollution and regional emission/stubble-burning inputs required by the chosen configuration.
Forecast	72-hour forecast generation.
Inversion	Inversion-strength indicator linked to forecast conditions.
Plume	Stubble-burning plume transport/impact visualization.
AQI	Spatial AQI and pollutant forecast products.
Dashboard	Map + timeline + pollutant + meteorology + inversion + plume views.
Validation	At least one historical event and a baseline comparison.

12. POST-MVP / DIFFERENTIATION

Enhancement	Value
Forecast uncertainty	Communicate confidence and uncertainty rather than a single deterministic value.
Advanced plume analytics	Estimate plume arrival, affected zones and expected contribution.
Scenario simulation	Explore how changes in emissions/weather conditions affect forecasts.
Automated alerts	Flag expected severe deterioration or strong inversion conditions.
Explainable drivers	Quantify the relative contribution of meteorological and chemical factors.
Longer historical evaluation	Demonstrate robustness across multiple pollution seasons/events.

13. OUT OF SCOPE / DO NOT CLAIM WITHOUT EVIDENCE

The PS does not prescribe a particular ML architecture, LLM, frontend framework, database or cloud provider. These should not be presented as requirements. Likewise, a dashboard alone, a conventional AQI regression model, or a mocked WRF-Chem output should not be represented as satisfying the coupled-modelling requirement.

14. KEY RISKS

Risk	Impact	Mitigation
WRF-Chem complexity	Very high	Start with a validated reference configuration and progressively adapt the domain.
Compute requirements	High	Benchmark small-domain runs early; plan offline/recorded runs for demo.
Data availability	High	Lock input sources and data contracts before UI development.
Validation difficulty	High	Define observations, baselines and metrics before claiming accuracy.
Over-scoped dashboard	Medium	Prioritize forecast, inversion, plume and pollutant views first.

15. SUCCESS CRITERIA

The product is considered SIH-ready when the team can demonstrate: **(1)** a genuine coupled weather–chemistry workflow, **(2)** a Delhi NCR 72-hour forecast, **(3)** spatial AQI/pollutant outputs, **(4)** inversion-strength tracking, **(5)** stubble-burning plume dispersion, **(6)** an operational dashboard, and **(7)** validation evidence supporting the forecast.

16. FINAL PRODUCT STATEMENT

SIH26082 product: A high-resolution Delhi NCR atmospheric forecasting and decision-support platform that couples meteorological and chemical processes to produce a 72-hour AQI outlook, explicitly models inversion and stubble-burning plume effects, and communicates the forecast through an actionable real-time dashboard.